File Systems & I/O Systems – MCQ Practice

30 Multiple-Choice Questions | Medium Difficulty

Instructions: Select the single best answer for each question.

1. A file system supports sequential access and direct access. Which operation is most appropriate when an application needs to read a specific record at a known position without reading all preceding records?

A. Sequential access
B. Direct access
C. Directory traversal
D. File allocation

2. In a tree-structured directory system, what is the main advantage of using absolute paths?

A. They always provide faster file access than relative paths
B. They identify a file location starting from the root directory
C. They eliminate file permissions
D. They allow files to exist without directory entries

3. Which problem is most closely associated with a single-level directory structure when many users store files?

A. Excessive page faults
B. File-name conflicts between different users' files
C. Lack of physical memory
D. Increased interrupt latency

4. Which file allocation method stores a file's blocks in consecutive disk locations?

A. Linked allocation
B. Indexed allocation
C. Contiguous allocation
D. Hashed allocation

5. A file system uses linked allocation. What is a key disadvantage of this method when accessing a block near the end of a file?

A. Every file must occupy contiguous disk blocks
B. Direct access is inefficient because block links must be followed
C. It requires a separate index block for every data block
D. It cannot support files that grow over time

6. What is the primary purpose of an Access Control List (ACL) associated with a file?

A. To specify the physical disk sectors occupied by the file
B. To define which users or groups have specific permissions on the file
C. To store the file's contents in memory
D. To determine the file's page-replacement policy

7. In a file system, a user has read permission but not write permission for a file. What should normally happen when the user attempts to modify its contents?

A. The write should be denied by the access-control mechanism
B. The file should automatically become read-write
C. The operating system should change the file owner
D. The file should be moved to the root directory

8. Which data structure typically stores metadata such as file size, ownership, permissions, and pointers to data blocks in a traditional Unix-like file system?

A. Process Control Block (PCB)
B. Inode
C. Translation Lookaside Buffer (TLB)
D. Device controller register

9. What is a major feature of the EXT4 file system?

A. It supports only read-only files
B. It uses extents to represent ranges of contiguous file blocks efficiently
C. It requires every file to occupy one contiguous disk region
D. It cannot maintain file metadata

10. In a typical Linux filesystem hierarchy, where are essential commands and utilities commonly located?

A. /dev
B. /proc
C. /bin and /usr/bin
D. /sys

11. What is a common reason for mounting the root filesystem read-only during the early stages of embedded Linux boot?

A. To prevent the kernel from accessing files
B. To allow initial filesystem checks or setup before enabling writes
C. To disable all device drivers
D. To prevent the system from loading its init process

12. Which statement best describes UBIFS?

A. A filesystem designed specifically for raw flash memory devices, typically used over UBI
B. A directory structure used only for network shares
C. A block allocation algorithm used by EXT4
D. A Linux command for mounting USB drives

13. What does the Linux mount operation do?

A. Formats a storage device automatically
B. Attaches a filesystem to a directory in the existing filesystem hierarchy
C. Changes a file's owner to root
D. Copies all files from the device into RAM

14. Which command is commonly used to display currently mounted filesystems and their mount points on Linux?

A. chmod
B. ps
C. findmnt
D. kill

15. What is the main role of a device controller?

A. To provide hardware-level control and manage communication with a device
B. To decide which user can access a file
C. To manage directory names and paths
D. To allocate virtual memory pages to applications

16. What is the primary role of a device driver in an operating system?

A. To replace the device controller's physical hardware
B. To provide software that allows the OS to communicate with and control a device
C. To store every device's data permanently in RAM
D. To schedule only user-level application threads

17. A device completes an I/O operation and signals the CPU through an interrupt. What is a key advantage of this mechanism over continuous polling?

A. The CPU must check the device status after every instruction
B. The CPU can perform other work until the device signals that attention is needed
C. The device no longer requires a driver
D. The operating system no longer needs to handle I/O errors

18. What is the primary purpose of Direct Memory Access (DMA)?

A. To allow a device to transfer data to or from memory with limited CPU involvement in the data transfer itself
B. To convert a filesystem from read-only to read-write
C. To replace virtual memory with physical addressing
D. To execute device drivers outside the operating system

19. Which statement best describes blocking I/O?

A. The I/O call returns immediately while the operation continues in the background
B. The calling thread waits until the I/O operation completes or otherwise reaches the relevant completion condition
C. The application never receives the I/O result
D. The kernel cannot schedule other threads while I/O is pending

20. An application uses non-blocking I/O to read from a socket, but no data is currently available. What typically happens?

A. The calling thread waits indefinitely
B. The operation returns promptly, indicating that it would block or that no data is currently available
C. The operating system terminates the socket
D. The kernel automatically converts the socket to blocking mode

21. What is the key distinction between synchronous and asynchronous I/O?

A. Synchronous I/O always uses interrupts, while asynchronous I/O always uses polling
B. Synchronous I/O relates completion to the calling operation, while asynchronous I/O allows the operation to proceed and completion to be reported later
C. Synchronous I/O can only be used with files, while asynchronous I/O can only be used with devices
D. There is no distinction between them

22. Which situation best illustrates polling?

A. A device notifies the CPU through an interrupt when an event occurs
B. The CPU repeatedly checks a device's status register to determine whether an operation has completed
C. A process waits for a child process to terminate
D. A filesystem updates its inode metadata

23. What is the purpose of the Linux kernel I/O subsystem?

A. To provide common mechanisms and abstractions for managing I/O across different devices
B. To replace all device-specific drivers with one universal hardware driver
C. To manage only user-space graphical applications
D. To eliminate the need for buffering and caching

24. What is the main purpose of the Linux device model?

A. To organize devices and their relationships in a consistent framework, supporting device discovery and driver interaction
B. To define the order in which files are stored on disk
C. To replace the Linux process scheduler
D. To ensure every device uses the same hardware interface

25. A server must monitor many file descriptors for readiness to perform I/O. Which statement about select(), poll(), and epoll() is correct on Linux?

A. select() and poll() are exclusively for regular disk files, while epoll() is only for terminals
B. epoll() is designed to scale efficiently for monitoring many file descriptors, particularly when only a subset is ready
C. select() automatically performs asynchronous reads for every descriptor
D. poll() can monitor only one file descriptor at a time

26. What is an important limitation of select() on many systems?

A. It cannot monitor sockets
B. It requires every monitored descriptor to be a regular file
C. It commonly has a fixed descriptor-set size limit and requires descriptor sets to be prepared again for each call
D. It always waits for every monitored descriptor to become ready

27. Why are device queues useful in an operating system's I/O subsystem?

A. They store file permissions permanently
B. They organize pending I/O requests for processing by a device or driver
C. They replace all hardware interrupts
D. They prevent multiple processes from opening files

28. A device queue contains several requests for a storage device. What can the operating system or driver do with these requests?

A. Schedule their processing according to the device's capabilities and applicable scheduling policy
B. Execute all requests simultaneously on the same hardware without constraints
C. Convert every request into a filesystem directory
D. Discard all requests from user processes

29. In embedded Linux, why might a system use a read-only root filesystem with selected writable locations?

A. To prevent the kernel from loading modules
B. To reduce the risk of unintended system-file modification and improve robustness against unexpected power loss
C. To guarantee that applications never generate errors
D. To eliminate the need for persistent storage

30. An application needs to wait until one of several sockets becomes ready without continuously checking each socket in a busy loop. Which approach is most appropriate?

A. Repeatedly call read() on every socket in a tight loop
B. Use an event-waiting mechanism such as poll() or epoll()
C. Remount the root filesystem as read-only
D. Use contiguous file allocation

Answer Key

Question	Answer	Question	Answer	Question	Answer
Q1	B	Q11	B	Q21	B
Q2	B	Q12	A	Q22	B
Q3	B	Q13	B	Q23	A
Q4	C	Q14	C	Q24	A
Q5	B	Q15	A	Q25	B
Q6	B	Q16	B	Q26	C
Q7	A	Q17	B	Q27	B
Q8	B	Q18	A	Q28	A
Q9	B	Q19	B	Q29	B
Q10	C	Q20	B	Q30	B